Comprehensive Career Development and Recruitment Framework using multi-causal AI

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Project Proposal Report

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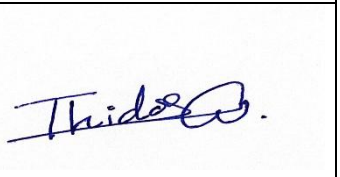
Sri Lanka Institute of Information Technology

Sri Lanka

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DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

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ABSTRACT

Things are changing fast with AI and automation taking over a lot in jobs around the world. Experienced workers might worry about whether their careers will hold up over time, especially with industries shifting so much. It feels like not many tools out there really help people in the middle of their careers deal with risks from machines replacing jobs or whole sectors becoming unstable. Most career advice systems I have seen are aimed at students or new grads. They match jobs based on skills or predict what career might fit, but they do not go deep into long term stuff for pros who have been working for a while. Like, automation risks or if skills can transfer to other areas, that kind of thing gets overlooked. Plus, a lot of these use fancy machine learning that spits out numbers without saying why or what to do next. It seems kind of frustrating for users who need real guidance. This study tries to fill that gap with a framework for analyzing career safety and how stable industries are. It looks at experienced professionals through a few main areas, automation exposure for one, then perceived stability in their field, skill transferability, and readiness to learn new things or reskills. Instead of black box models, it uses rules to classify risks and explain what is going on, which makes it more transparent. The idea is to gather info from surveys on these dimensions and process it with a simple decision engine. That way, it categorizes the risk level and gives personal tips on handling potential problems. I think integrating all these factors helps show a fuller picture of vulnerabilities and what strengths someone has. For building it, the system would use web tech like React for the front-end where users interact, Node.js for the back-end, and Python for model training, something like MongoDB to store responses and results. The rule-based part handles the analysis to output the classifications and advice. In the end, the goal is a tool that explains career risks clearly so professionals can spot threats early and plan ways to stay resilient. On a bigger scale, it could help the workforce adapt better in an economy full of AI, maybe push for ongoing learning. Businesses might plug it into HR setups or counseling sites to help with decisions on careers. Some people might see it as just another app, but others could find it useful for real planning.

Keywords: career risk analysis, automation exposure, industry stability, explainable AI, career development.

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LIST OF ABBREVIATIONS

Abbreviation	Full Meaning
AI	Artificial Intelligence
ML	Machine Learning
API	Application Programming Interface
UI	User Interface
UX	User Experience
DB	Database
IT	Information Technology
SDG	Sustainable Development Goals
KPI	Key Performance Indicator
NLP	Natural Language Processing
CSV	Comma-Separated Values
JSON	JavaScript Object Notation
WBS	Work Breakdown Structure
API	Application Programming Interface
ML Model	Machine Learning Model

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1. INTRODUCTION

1.1 Background and Context

Technology is changing jobs so fast with stuff like AI and automation, and its making people worry about their work. A lot of old-school jobs might get taken over or changed completely, especially if they are routine ones. I think experienced workers in those fields are starting to wonder if what they know now will even matter down the line.

Mid-career folks seem to be hit hardest by this. They are not like new grads picking paths for the first time. These people have built up their roles over the years, and now tech could shake that up. Without good info on risks or how to switch skills, it's tough for them to plan ahead or decide what to do next.

Reports from places like the World Economic Forum say millions of jobs could shift because of automation in the next few years. Industries are flipping around quickly, too, so learning new things and being flexible matter more than ever. At the same time, current career tools mostly help students or people hunting for jobs, like with resumes or matching skills to openings.

Those platforms use machine learning to guess fits, but they do not really look at long-term safety for pros with experience. Plus, the results come out as numbers without much why, so users might not get it. Some people might find that frustrating, I guess.

To fix this, the idea here is a framework for checking career safety and how stable industries are, aimed at experienced workers. It looks at things like how much automation hits a job, if skills can move around, and readiness to adapt. Using rules instead of just black box models, it explains risks clearly and suggests steps to lower them.

This ties into UN goals for good work and growth, pushing for jobs that last in tough economies. The system fits in AI research, especially explainable parts, decision tools, and career analysis. That part gets a bit messy to explain fully, but it seems important for helping people stay ahead.

Table 1: Career Risk Indicators Used in the System

Indicator	Description
Automation Exposure	Measures the likelihood that a job role may be affected by AI or automation.
Industry Stability	Evaluates whether the industry is expected to remain stable in the next five years.
Skill Transferability	Determines whether current skills can be easily applied to other job roles.
Adaptability Readiness	Measures the willingness of employees to learn new skills or reskill.

1.2 Literature Review

Recent studies have explored how artificial intelligence and data analytics can support career guidance and workforce trend analysis. Many of these studies focus on applying machine learning algorithms to predict career suitability or to match individuals with appropriate job roles based on their skills and interests. For example, Bannaka et al. (2021) proposed a machine learning-based framework that predicts career fit, career progression, and potential employee turnover, primarily targeting information technology graduates. Their model evaluates factors such as academic performance, personality traits, and career preferences to recommend suitable career paths. Although the approach demonstrates the effectiveness of predictive analytics for career planning, it mainly focuses on students and early-career individuals rather than experienced professionals already working in different industries [1].

Similarly, other studies have investigated AI-based job recommendation systems that utilize supervised learning techniques to match candidate profiles with job descriptions. These systems analyze skill sets, educational background, and personal interests to suggest potential career opportunities. For instance, research by Paparrizos et al. highlighted how machine learning models can analyze large-scale employment datasets to improve job matching and recruitment processes. However, these systems primarily concentrate on recommending job opportunities and often overlook broader factors such as long-term industry stability, technological disruption, or automation risks that may influence career sustainability . [2]

In addition to academic research, several commercial career platforms provide online skill assessments and automated job recommendations. These systems aim to assist individuals in finding employment opportunities quickly by evaluating their competencies and matching them with available job roles. While such platforms are widely used, many operate as black-box models, providing predictions without clear explanations of the underlying decision-making process. As noted by Bogen and Rieke (2018), the lack of transparency in algorithmic decision systems can make it difficult for users to fully understand or trust the recommendations provided by automated employment tools [3].

Although these approaches provide valuable tools for career guidance and job matching, there remains a limited focus on analyzing long-term career sustainability and industry stability, particularly for experienced professionals. Rapid technological advancements, digital transformation, and the increasing adoption of automation technologies are continuously reshaping labor markets and altering the demand for specific skills. According to Frey and Osborne (2017), a significant portion of current occupations face potential automation risks, emphasizing the need for systems that analyze not only job suitability but also future career stability [4]. Therefore, there is a growing need for intelligent systems that combine workforce data, industry trends, and predictive analytics to provide more comprehensive insights into career safety and long-term employment sustainability.

1.3 Research Gap

Even though research on career stability and workforce mobility has advanced significantly, several important research gaps remain in the area of career safety and industry stability analysis. Many existing studies mainly focus on analyzing employment trajectories and workforce transitions rather than providing predictive systems that support career decision-making. For instance, the study by Ojala et al. analyzed career stability among Finnish industrial employees between 1988 and 2015 by examining employment transitions and workforce mobility patterns across multiple cohorts [5]. However, such studies primarily rely on historical employment data and statistical analysis rather than developing intelligent systems that can evaluate or predict future career safety or industry stability for individuals.

Another limitation in current research is the strong reliance on traditional statistical and regression-based approaches to analyze career patterns. Studies such as Rieckhoff et al. investigated mid-career stability by examining labor market turbulence indicators and employment transitions across cohorts [6]. While these approaches provide valuable insights into workforce trends and job stability patterns, they do not offer predictive decision-support tools that help students, graduates, or job seekers evaluate the long-term stability of different industries or career paths.

Furthermore, existing research highlights that career stability is influenced by multiple factors such as economic fluctuations, technological changes, industry demand, and labor market dynamics. Research examining the relationship between career stability and earnings mobility also indicates that employment transitions, tenure, and job changes can significantly affect long-term career outcomes [7] [8] [9] [10]. However, most existing studies analyze these factors separately and rarely integrate them into a unified analytical framework capable of predicting career safety or industry stability.

Another major gap is the lack of AI-driven career analysis systems that combine labor market data, industry trends, and predictive modeling to provide real-time career insights. Current research primarily focuses on descriptive analysis of workforce patterns rather than developing intelligent decision-support frameworks that can assist individuals in selecting stable and sustainable career paths.

The research gap, therefore, focuses on (i) the absence of predictive AI-based systems for career safety analysis, (ii) the dependence on historical statistical analysis instead of intelligent forecasting, (iii) the lack of integrated frameworks combining labor market indicators and industry trends, and (iv) the limited availability of data-driven tools that support career decision-making. The proposed Career Safety and Industry Stability Analysis framework aims to address these limitations by utilizing machine learning techniques and multi-factor labor market analytics to provide predictive insights into career stability and industry risk.

Table 2 : Research gap of career safety and risk analysis

Study / Research	Method Used	Focus Area	Limitations	Gap Addressed by Proposed System
Miika Ojala et al. (2017)	Statistical analysis of employment records	Career stability and workforce mobility among Finnish industrial employees	Use historical employment data only and analyzes past workforce patterns	No predictive system to forecast career safety or industry stability
Benjamin Rieckhoff et al. (2020)	Labor market analysis using regression models	Mid-career stability and labor market turbulence	Focuses on statistical indicators rather than predictive tools	Does not support career decision-making for individuals
Career Stability & Earnings Mobility Studies	Economic and labor market analysis	Relationship between job transitions, tenure, and earnings mobility	Factors analyzed separately without integrated analysis	Lack of unified framework combining multiple labor market factors
Industry Demand and Workforce Trend Studies	Descriptive workforce trend analysis	Impact of technology, economic changes, and industry demand	Mostly descriptive analysis of workforce data	No AI-driven framework for real-time career insights
Proposed System: Career Safety & Industry Stability Analysis Framework	Machine Learning + Multi-factor Labor Market Analytics	Predictive career safety evaluation and industry stability analysis	Overcomes limitations of traditional research	Provides AI-based predictions and decision-support for career planning

2. OBJECTIVE

2.1 Main Objective

The main objective of this research is to develop an AI-based Career Safety and Industry Stability Analysis system that evaluates and predicts the stability and risk level of different career paths using labor market data, industry trends, and machine learning techniques to support informed career decision-making.

2.2 Specific Objectives

- To collect and analyze labor market and industry data from relevant sources to identify key factors that influence career stability and employment security.
- To design a multi-factor career stability evaluation model that analyzes variables such as industry growth, job demand, technological disruption, and employment trends.
- To develop a machine learning model capable of predicting career safety levels and industry stability based on historical and real-time labor market data.
- To implement a data-driven analytical framework that integrates industry indicators, employment statistics, and predictive algorithms for career stability assessment.
- To evaluate the performance and accuracy of the proposed model using relevant datasets and validation techniques.
- To provide a decision-support system for students and job seekers that generates insights and recommendations regarding stable and sustainable career paths.

3. METHODOLOGY

3.1 Research Approach

This research follows a Design Science Research (DSR) approach, which focuses on designing and developing an innovative technological solution to solve a real-world problem. In this study, the problem addressed is the lack of intelligent systems capable of evaluating career safety and industry stability for students and job seekers.

The research involves designing and implementing an AI-driven analytical framework that integrates labor market data, industry indicators, and machine learning models to predict career stability. The approach includes system design, model development, experimentation, and evaluation of the predictive performance of the proposed solution.

3.2 Technical Methods and Algorithms

The proposed system will utilize several data analysis and machine learning techniques to evaluate career safety and industry stability.

First, data preprocessing techniques such as data cleaning, normalization, and feature selection will be applied to prepare labor market and industry datasets.

Second, clustering algorithms such as K-Means Clustering will be used to group industries based on stability indicators like job demand, employment trends, and market growth.

Third, machine learning classification models such as Random Forest and Logistic Regression will be used to predict the stability level of career paths based on multiple influencing factors.

Additionally, data visualization techniques will be used to present insights about industry stability and career safety trends in a user-friendly manner.

3.3 Tools and Platforms with Justification

Several tools and technologies will be used to develop and evaluate the proposed system.

- **Python** will be used as the primary programming language because of its strong ecosystem for machine learning and data analysis.
- **Scikit-learn** will be used to implement machine learning algorithms such as clustering and classification models.
- **Pandas and NumPy** will be used for data processing and analysis of large datasets.
- **Matplotlib or Seaborn** will be used for data visualization to present insights about career stability.
- **Jupyter Notebook** will be used as the development environment for experimentation and model evaluation.

3.4 Data Collection and Ethical Considerations

The system will use secondary datasets related to labor market trends, employment statistics, and industry growth indicators. These datasets may be obtained from publicly available sources such as government labor statistics, employment databases, and global workforce reports

During the data collection process, ethical considerations will be maintained by ensuring that no personal or sensitive information is collected without consent. Only publicly available and anonymized datasets will be used for analysis. Data will be handled responsibly to maintain privacy and confidentiality.

3.5 Validation Strategy and Performance Metrics

To evaluate the effectiveness of the proposed system, the developed machine learning models will be tested using standard model validation techniques such as train-test split and cross-validation.

The performance of the models will be evaluated using several evaluation metrics, including:

- Accuracy – to measure the correctness of predictions
- Precision and Recall – to evaluate prediction quality for career stability classification
- F1-Score – to balance precision and recall for better model evaluation
- Confusion Matrix – to analyze prediction performance across different classes

These metrics will help determine the reliability and predictive capability of the proposed Career Safety and Industry Stability Analysis framework.

4. HIGH-LEVEL SYSTEM ARCHITECTURE

The proposed Career Safety and Industry Stability Analysis System is designed as a modular architecture that integrates data collection, machine learning analysis, and user interaction components to provide meaningful career insights. As illustrated in the system architecture diagram, the solution begins with multiple data sources, including employee survey responses, industry reports, labor market datasets, and online job trend platforms. These inputs are collected through web forms, APIs, and external databases. The collected raw data is first stored in a centralized data storage layer, where preprocessing operations such as data cleaning, transformation, and feature extraction are performed to ensure data quality and consistency for analysis.

After preprocessing, the data is passed to the analytics and machine learning layer, which represents the core intelligence of the system. This layer includes algorithms responsible for analyzing career trends, identifying industry stability patterns, and predicting potential risks related to layoffs, automation impact, and skill relevance. The models process the prepared datasets and generate predictive insights related to career safety and employment stability. The results produced by the machine learning models are then stored in structured databases to support further analysis and retrieval.

The processed outputs are delivered to the application and visualization layer, which consists of the user interface and analytical dashboard. This component allows users such as students, employees, career advisors, and researchers to interact with the system through a web-based platform. Users can submit career-related data, view industry stability scores, analyze skill demand trends, and obtain recommendations for reskilling or career transitions. The interaction between components follows a continuous data flow where user inputs and survey data are collected, processed, analyzed by the predictive models, and finally visualized through dashboards and reports.

The architecture is designed with scalability and security considerations to ensure reliable system performance. The use of modular components and API-based integrations allows the system to expand by incorporating additional datasets or analytical models in the future. Secure data handling mechanisms, authentication processes, and controlled database access ensure that sensitive employee and survey information is protected. Overall, this high-level architecture enables an efficient and scalable framework for analyzing career safety trends and supporting data-driven career decision-making.

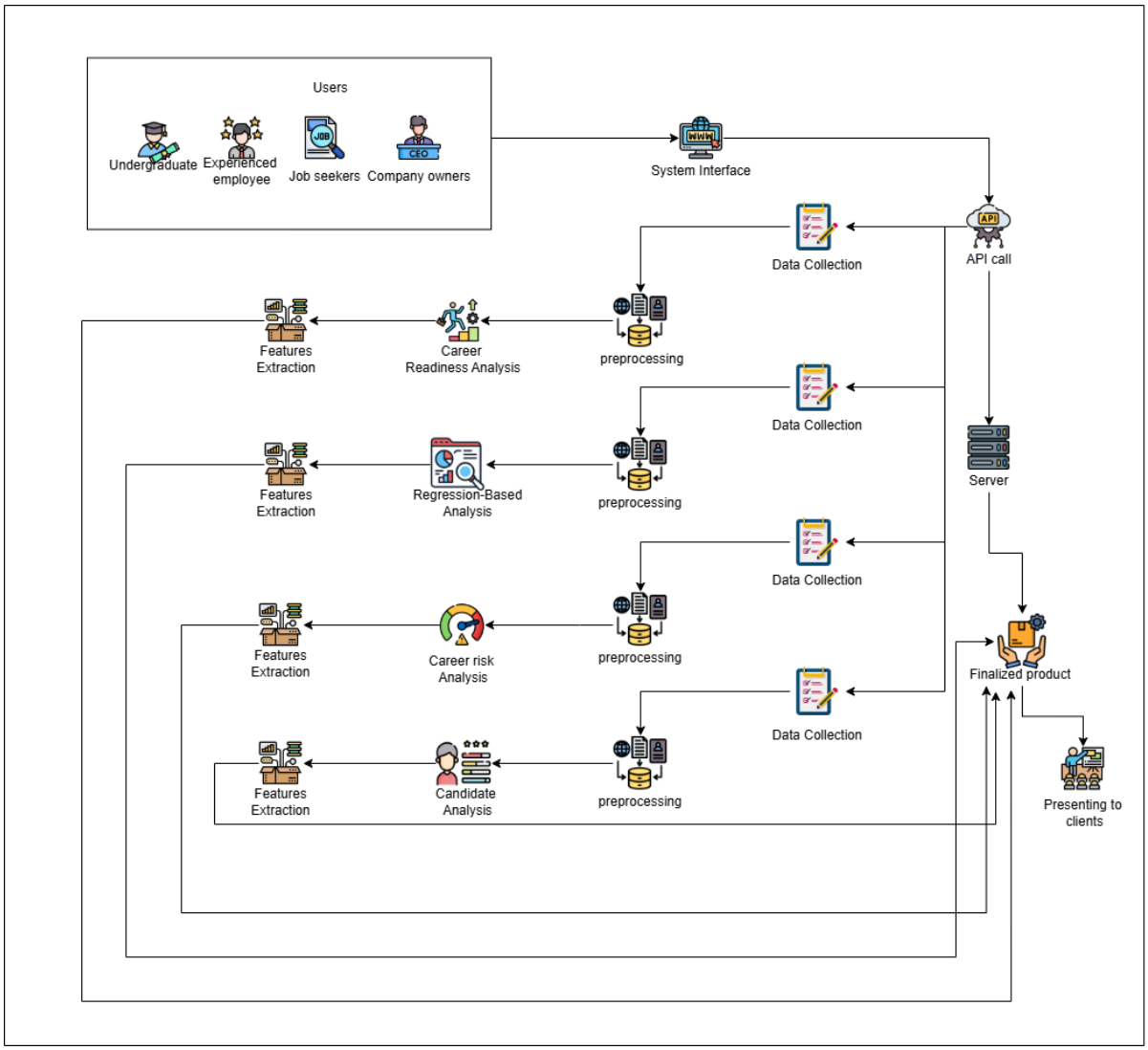


Figure 1: Overall system diagram

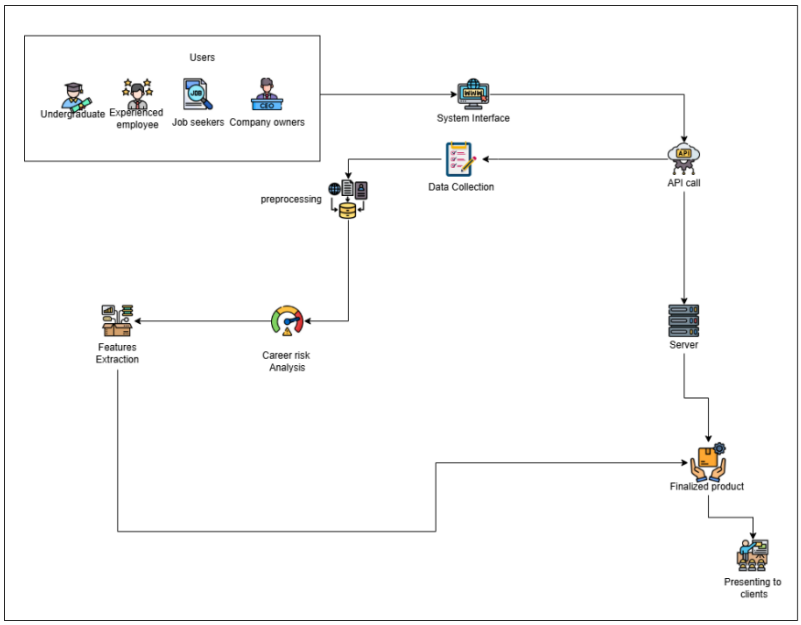


Figure 2 : Career Safety and Industry Stability Analysis System Workflow

5. USER REQUIREMENTS

The proposed Career Safety and Industry Stability Analysis system involves several stakeholders who interact with or benefit from the system. The primary stakeholders are students, graduates, and job seekers who require reliable insights about career stability and industry trends to make informed career decisions. These users will utilize the system to evaluate the safety and long-term stability of different career paths based on industry data and predictive analysis.

Another important group of stakeholders includes career counselors and academic institutions, who can use the system as a decision-support tool when guiding students in selecting suitable career paths. By analyzing industry demand and employment trends, counselors can provide better recommendations to students regarding stable and sustainable careers.

Additionally, researchers and policymakers may benefit from the analytical insights generated by the system. The system can provide valuable information about workforce trends and industry stability, which may support workforce planning and labor market policy development. Finally, system developers and administrators are also stakeholders responsible for maintaining the system, managing datasets, and ensuring the proper functioning and performance of the platform.

5.1 Functional Requirements

The functional requirements describe the core functionalities that the proposed system must perform in order to achieve the research objectives.

1. The system shall collect and process labor market and industry-related datasets from relevant sources.
2. The system shall allow users to input career preferences, industry selections, or related parameters for analysis.
3. The system should analyze industry trends and employment data to evaluate the stability of different industries.
4. The system shall apply machine learning algorithms to predict the stability level of specific career paths.
5. The system shall categorize career options into different stability levels such as high stability, moderate stability, and high risk.
6. The system shall generate analytical reports and visualizations that present career safety insights and industry trends.
7. The system shall provide recommendations or insights to assist users in selecting stable career paths.
8. The system shall store and manage processed data and analytical results for future reference and analysis.

5.2 Non-Functional Requirements

Non-functional requirements describe the quality attributes and operational constraints that the system must satisfy in order to perform effectively and reliably.

The system must ensure security by protecting the stored data and preventing unauthorized access to the system. Appropriate authentication and data protection mechanisms should be implemented to maintain the integrity and confidentiality of the information used within the platform.

The system should maintain a high level of performance by processing career analysis requests efficiently and providing results within an acceptable response time. Efficient algorithms and optimized data processing techniques will be required to handle large datasets related to labor market trends.

In terms of usability, the system should provide a user-friendly interface that allows students and job seekers to easily interact with the platform and interpret the analytical results. The interface should present career stability insights through clear visualizations and understandable reports.

The system should also maintain reliability and scalability, ensuring that the platform operates consistently and can handle increasing amounts of data or users in the future. Additionally, the system should be designed with maintainability in mind, allowing developers to update algorithms, datasets, and system features when necessary.

6. COMMERCIALIZATION PLAN

Phase 1: Pilot Implementation (Free Access for Universities and Career Counselors)

In the first phase, the system will be introduced through pilot implementations in collaboration with undergraduate students and IT companies. During this stage, the platform will be provided free of charge to selected universities, career counselors, and research groups to evaluate its effectiveness in real-world environments. Students and career advisors will be able to analyze career paths and industry stability using the system's predictive analytics features. The main objective of this phase is to test the AI models, improve prediction accuracy, and identify practical challenges that may arise when using labor market data and career trend analysis. Feedback collected from students, counselors, and researchers will be used to refine the machine learning models, improve the user interface, and enhance the overall reliability of the system before large-scale deployment.

Phase 2: Public Platform with Freemium Access

In the second phase, the system will be launched as a public web-based platform that allows students, graduates, and job seekers to access career stability insights. Users will be able to enter career preferences or industry selections and receive analysis on career safety levels, industry growth trends, and long-term stability predictions. A freemium pricing model will be introduced at this stage. The free version will allow users to perform a limited number of career analyses per day and access basic insights about industry stability. A premium subscription tier will provide additional features such as detailed career stability reports, advanced predictive analytics, unlimited analysis requests, and personalized career recommendations. This phase will also allow the system to collect more data about user interactions and industry patterns, which will help improve the predictive performance of the machine learning models.

Phase 3: Enterprise Integration and Institutional Licensing

The final phase of commercialization will focus on expanding the system to enterprise and institutional customers, including universities, recruitment agencies, career counseling organizations, and human resource departments of companies. In this stage, organizations will be able to integrate the system into their platforms through subscription-based licensing or API integration. For example, universities can use the system to support student career guidance programs, while recruitment platforms may integrate career stability analytics to improve job matching processes. Enterprise clients will be able to access advanced analytical dashboards, workforce trend reports, and industry stability insights. This phase will generate sustainable revenue while positioning the platform as a trusted AI-driven career intelligence solution that supports workforce planning, career development, and labor market analysis.

7. BUDGET AND JUSTIFICATION

Table 3: Budget and justification

Description	Amount
Deployment and hosting	LKR 20,000
Cloud Services	LKR 5,000
Training and Database	LKR 40,000 (Use of free resources will be attempted)
Travelling	LKR 4,000

8. WORK BREAKDOWN STRUCTURE (WBS)

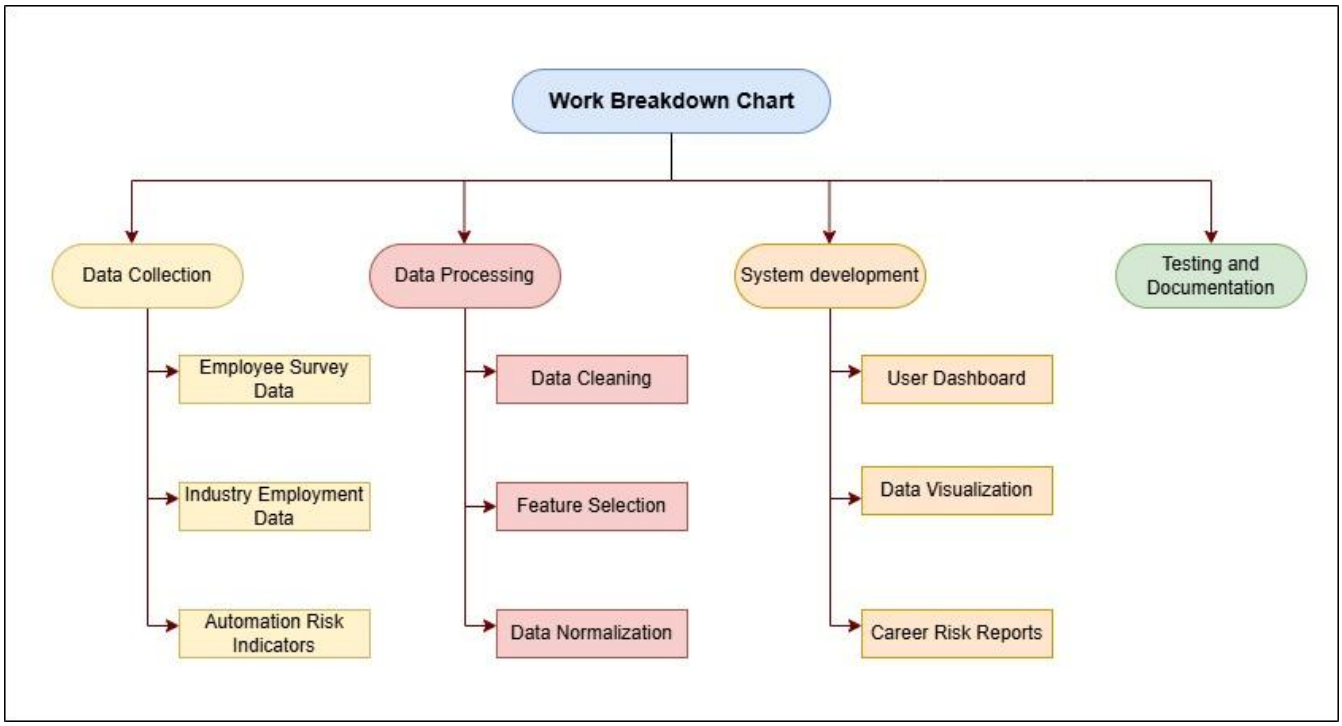


Figure 3: Work breakdown chart

9. GANTT CHART

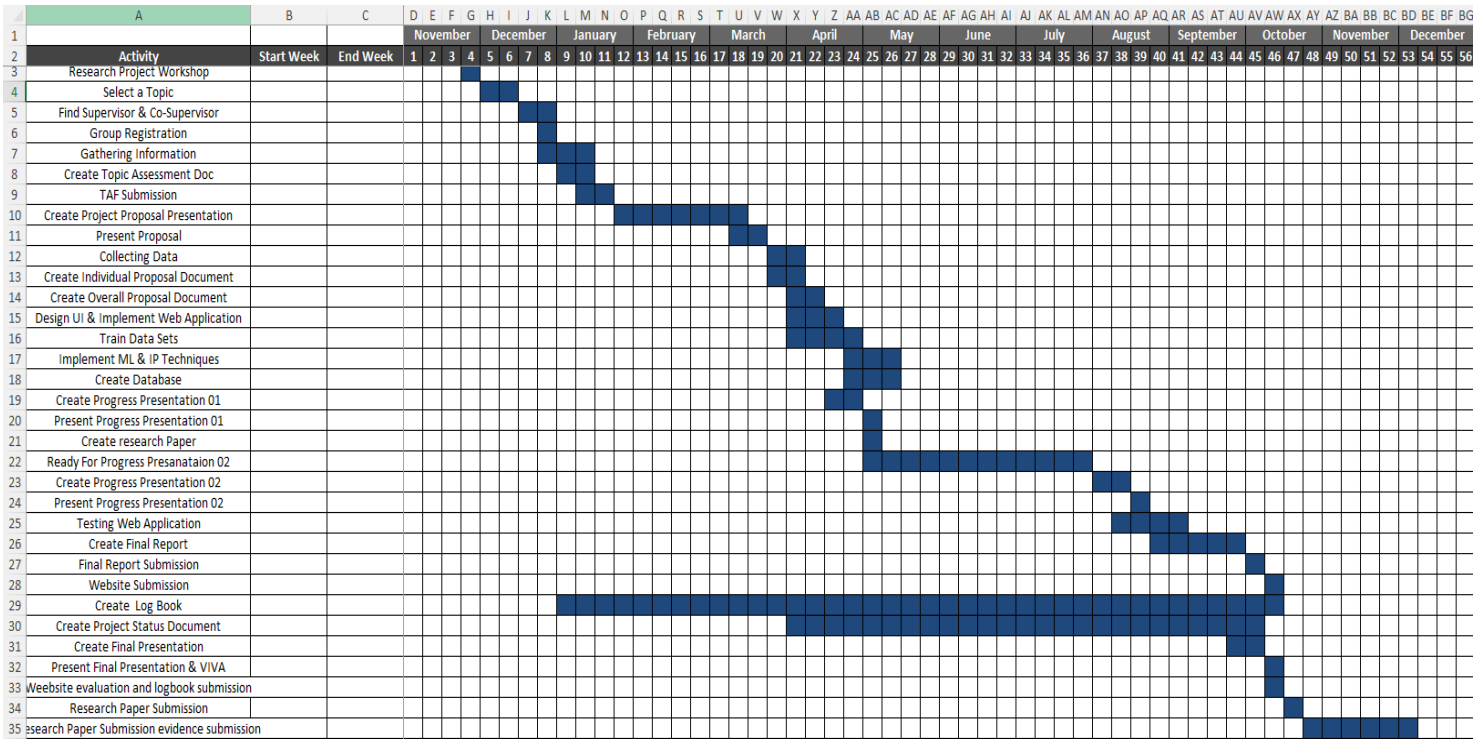


Figure 4:Grantt chart

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11. APPENDICES

Appendix A – Employee Survey Questionnaire

This appendix contains the survey instrument designed to collect data from employees regarding career stability, skill relevance, automation concerns, and industry trends. The responses gathered from this questionnaire will support the analysis of career safety factors and provide data for validating the proposed machine learning model.

Survey Questions

1. Years of Work Experience
    - Less than 1 year
    - 1–3 years
    - 3–5 years
    - 5–10 years
    - More than 10 years
2. Current Job Role / Position
3. Which industry do you currently work in?
    - IT Support Engineer
    - Software Engineer
    - Quality Engineer
    - AI/ML Engineer
    - Project Manager
4. How relevant are your current skills to your job role?
    - Very relevant
    - Moderately relevant
    - Slightly relevant
    - Not relevant
5. How stable do you believe your industry is?
    - Very stable
    - Moderately stable
    - Uncertain
    - Unstable
6. How concerned are you about job loss due to automation or AI?
    - Very concerned
    - Moderately concerned

- Slightly concerned
- Not concerned

7. Have you experienced layoffs or job insecurity during your career?

    - Yes
    - No

8. Have you recently taken steps to reskill or learn new technologies?

    - Yes
    - No

9. How confident are you about your long-term career stability?

    - Very confident
    - Somewhat confident
    - Neutral
    - Not confident

Appendix B – Preliminary UI Mockups

This appendix presents the preliminary user interface mockups of the proposed system. These mockups illustrate how users will interact with the platform to evaluate career safety and industry stability. The interface is designed to provide a simple and user-friendly experience for employees who want to analyze their career risk level and industry trends.

The main interface components include the **dashboard, career risk analysis page, skill relevance analysis page, and industry stability visualization panel**. The dashboard will display key indicators such as career stability score, automation risk level, and recommended reskilling options. Users will be able to input their job role, years of experience, and skill set in order to generate a career safety prediction using the machine learning model.

The UI design focuses on clarity and accessibility, ensuring that users can easily understand the analytical results produced by the system. These mockups serve as a preliminary visualization of the final system interface and will guide future system development

Appendix C – Risk Analysis

This appendix identifies potential risks that may affect the development and implementation of the proposed career safety prediction system. One potential risk is the **limited availability of high-quality workforce and industry datasets**, which may impact the accuracy of the machine learning model. Another risk involves **bias in the training data**, which could affect prediction fairness across different industries or job roles.

Technical risks may also arise during system development, such as integration issues between system components or performance limitations when processing large datasets. Additionally, user adoption may become a challenge if the system results are not easily interpretable or trusted by users.

To mitigate these risks, the project will implement strategies such as using reliable public datasets, applying data preprocessing and validation techniques, and continuously evaluating the machine learning model performance. Regular testing and feedback from potential users will also help improve system reliability and usability.

Appendix D – Ethical Considerations

This appendix outlines the ethical considerations related to the collection and use of employee data for the proposed research. Participation in the survey will be **completely voluntary**, and respondents will be informed about the purpose of the research before providing their responses.

All collected data will be **anonymized to protect participant privacy**, and no personally identifiable information will be stored or published. The data will be used strictly for academic research and analysis purposes related to the development of the career safety prediction system.

Furthermore, the research will follow responsible AI practices to ensure that the system does not promote discrimination or unfair bias when predicting career stability. Ethical guidelines will be followed throughout the research process to ensure transparency, privacy protection, and responsible use of data.